

Lesson 1: Mini-project: Clean and visualize the data from your coffee track

=== LESSON ===

=== Concept ===

Alright! Today, we're going to dive into cleaning and visualizing data from our coffee tracking project. Think of it like organizing your pantry - you have all these bags of different types of coffee beans, roast levels, and brewing methods, but to truly enjoy them, they need to be neatly sorted and displayed. Similarly, our data needs to be cleaned (sorted) and visualized (displayed), so we can better understand its story.

=== Step-by-Step Walkthrough ===

Let's start by loading the necessary libraries and reading in our data. We'll use pandas for data manipulation, seaborn for visualization, and matplotlib for more customizable plots.

```
```python
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

Load the coffee data
coffee_data = pd.read_csv('coffee_data.csv')
```
```

Now, let's clean the data by removing any missing values:

```
```python
Remove any rows with missing values
```

```
coffee_data.dropna(inplace=True)
```

```
'''
```

Next, let's explore the data to understand what we have:

```
```python
```

```
# Display the first 5 rows of the data
```

```
print(coffee_data.head())
```

```
'''
```

Now, let's visualize the distribution of coffee types by using a bar plot:

```
```python
```

```
Create a bar plot of coffee types
```

```
sns.countplot(x='Coffee Type', data=coffee_data)
```

```
plt.show()
```

```
'''
```

Finally, let's visualize the average rating for each coffee type using a box plot:

```
```python
```

```
# Create a box plot of average ratings for each coffee type
```

```
sns.boxplot(x='Coffee Type', y='Rating', data=coffee_data)
```

```
plt.show()
```

```
'''
```

=== Your Mini Project Challenge ===

Using the code above as a starting point, add more visualizations to your project. For example, you could create a scatter plot showing the relationship between brewing method and taste intensity or a heatmap of the distribution of ratings across all coffees.

Remember, the goal is to understand the data better by exploring various visual representations. Happy exploring!

=== What Success Looks Like ===

Success might look like a collection of clean, informative, and visually appealing plots that help you understand your coffee data better. For example:

- A bar plot showing the distribution of coffees by type.
- A box plot comparing average ratings across different coffee types.
- A scatter plot or heatmap to explore relationships between various attributes like brewing method, taste intensity, and acidity level.

Keep in mind that every project is unique, so your plots may look a bit different depending on the data you have. Regardless, always aim to learn something new from your visualizations!

=== COACH FEEDBACK ===

=== What You Did Well ===

Firstly, congratulations on completing your mini-project! It's fantastic that you managed to clean and visualize data from your coffee tracking project. Here are some areas where you did well:

1. Loading the Data: You successfully loaded your data using a pandas DataFrame, which is an efficient way to handle tabular data in Python.

2. Cleaning the Data: You removed duplicates, filled missing values, and converted some column types, demonstrating a good understanding of data cleaning principles.
3. Visualizing the Data: You created a bar plot using matplotlib to visualize your coffee consumption over time. This is an excellent way to gain insights from your data!

=== Fixes & Suggestions ===

1. Error Handling: While working with data, it's essential to anticipate potential errors and handle them gracefully. Consider adding try-except blocks for loading or processing the data to ensure your code doesn't break in unexpected situations.
2. Code Organization: To make your code more maintainable and easier to understand, you could split your code into functions. This would allow you to reuse the cleaning steps for future projects and make your script more modular.
3. Documentation: It's essential to document your code so that others can understand what it does. Adding comments explaining each step of your data processing pipeline would be helpful.

=== Pro Tips to Go Further ===

1. Exploratory Data Analysis (EDA): To gain deeper insights into your coffee consumption, consider performing additional EDA steps, such as calculating summary statistics or creating correlation plots. This could help you identify patterns or relationships in your data that might not be immediately apparent from the bar plot alone.
2. Feature Engineering: Consider creating new features based on your existing data to uncover hidden insights. For example, you could create a 'coffee per day' column by dividing the total consumption by the number of days.
3. Data Validation: To ensure the quality of your data, consider using methods such as outlier detection or regression diagnostics. This can help identify potential errors in your data and improve

the accuracy of any models you build on this data.

Keep up the great work! You're well on your way to becoming a data analysis pro. Remember to always ask questions, seek feedback, and keep learning you've got this!